CHROMATIC NULLSTELLENSATZ LECTURE 1

TYPED BY AOWEN FAN

In the first lecture given by Guchuan Li, we begin with some basic setups for chromatic homotopy theory. We then explain the main results of [BSY22], the idea behind them and their applications without presenting any proof.

1. Chromatic homotopy theory

In the chromatic world, we fix a prime p and a height $n \in \mathbb{N}$. The central object is the Lubin-Tate theory.

Theorem 1.1 ([GH04]; [Rez98])**.** *Given a pair $(k, \mathbb{G})$, where k is a perfect field of character p and $\mathbb{G}$ is a formal group of height n, there is an $\mathbb{E}_\infty$-ring $E_n(k, \mathbb{G}) \in \mathrm{CAlg}(\mathrm{Sp}_{K(n)})$, such that $E_n(k, \mathbb{G})$ is even periodic and $\pi_0 E_n(k, \mathbb{G})$ can be identified with the Lubin-Tate ring.*

Throughout this seminar, we focus on an algebraic closed field $k = \overline{k}$, in which case there is only one formal group up to isomorphism. Applying the above construction, we obtain $E_n(k)$ called **height n Lubin-Tate** theory. The homotopy group can be characterized as

$$\pi_*(E_n(k)) = W(k)[\![u_1, \cdots, u_n]\!][v^{\pm 1}]$$

where $W(k)$ is the Witt ring of k, $u_1, \cdots, u_n$ are in degree 0 and v is in degree -2.

Another class of spectra is also of great interest, which is denoted by $K(n)$ and is called **Morava K-theory**, with homotopy ring given by

$$\pi_*(K(n)) = \mathbb{F}_p[v_n^{\pm 1}] \text{ where } |v_n| = (2p^n - 2)$$

It admits a structure of "field of ring spectra". But it has merely an $\mathbb{E}_1$-structure. One can work with $K(n)$-local category $\mathrm{Sp}_{K(n)}$. Especially, one has

$$\mathbb{S}_{K(n)} \simeq E_n(\overline{\mathbb{F}}_p)^{hG_n}$$

where G_n denotes the Morava stablizer group.

2. $T(n)$-local category

We now define $\mathrm{Sp}_{T(n)}$, which is the Bousfield localization of the " finite type-n-complex".

Proposition 2.1. *Let X be a finite p-local spectrum. Then $K(n)_*X = 0$ implies $K(n-1)_*X = 0$.*

Proof. See for example [Lur25][§26; Lemma 4]. □

Say a spectrum X is of type n if $K(n)_*X = 0$ and $K(n-1)_*X \neq 0$. It exists(also see [Lur25][§26]) and the $T(n)$-local category $\mathrm{Sp}_{T(n)}$ is defined as the X-local category, which does not rely on the choice of X.

Date: January 20, 2026.

3. The Nullstellensatz

The classical Hilbert Nullstellensatz says that suppose k is algebraic closed and $f_1, \cdots, f_n \in k[x_1, \cdots, x_m]$, if the equation system $f_i = 0$ has a solution, then there exists $g_1, \cdots, g_n$ such that $g_1 f_1 + \cdots + g_n f_n = 1$. This can generalize to abstract settings.

Definition 3.1. (Nullstellensatz) Let $\mathcal{C}$ be a presentable category. We say an object $k \in \mathcal{C}$ is **nullstellenstatz** if for any compact nonterminal object $A \in \mathcal{C}_{k/-}$, the map $k \to A$ admits a retract.

Back to $k[x_1, \cdots x_n]$, if we consider the polynomial function system $(f_1, \cdots, f_n)$ with a solution and let $A = k[x_1, \cdots x_m]/(f_1, \cdots, f_n)$. The existence of solution guarantees that we can find a retraction $k \to A \to k$.

The main results can be stated as follows.

Theorem 3.2 (Theorem A)**.** *Let $\mathcal{C} = \mathrm{CAlg}(\mathrm{Sp}_{T(n)})$. Then the nullstellensatz objects are of the form $E(L)$, the Lubin-Tate theory attached to an algebraic closed field L.*

Theorem 3.3 (Theorem D)**.** *Suppose R is a $T(n)$-local $\mathbb{E}_\infty$-ring, then there exists $R \to E(L)$ for some L.*

This immediately proves a theorem of Hahn.

Theorem 3.4 ([Hah16])**.** *$R \in \mathrm{CAlg}(\mathrm{Sp}_p)$, then $R \otimes T(n) = 0$ implies $R \otimes T(n+1) = 0$. Similarly for $K(n)$.*

Hahn's result makes it possible for us to define the height of $R \in \mathrm{CAlg}(\mathrm{Sp})$. We say that R is of height n if $R \otimes T(n) \neq 0$ but $R \otimes T(n+1) = 0$.

Theorem 3.5 (Redshift)**.** *Suppose $R \in \mathrm{CAlg}(\mathrm{Sp})$ is nonzero, then $\mathrm{ht}(K(R)) = \mathrm{ht}(R) + 1$.*

The proof can reduce to the case where $R = E(L)$, which reduces to a theorem of [Yua24] that states $\mathrm{ht}(K(E(L))) = \mathrm{ht}(E(L)) + 1$.

4. $\mathbb{E}_\infty$-complex orientation

Question: Does the complex orientation $\mathrm{MU} \to E_n(L)$ admit $E_n(L)$ an $\mathbb{E}_\infty$-complex orientation?

In order to solve this, we will need to construct the so called univeral orientability. The answer to the question in this paper is

Theorem 4.1 (Theorem G)**.** *There exists an $\mathbb{E}_\infty$-complex orientation $\mathrm{MU} \to E_n(\overline{\mathbb{F}}_p)$.*

Theorem D cannot complete the proof of the above theorem. Since it only makes sure the existence of some extension $R \to E_n(L)$ for some unpredicted L. Therefore, we will need the following variant.

Theorem 4.2 (Theorem F)**.** *Fix L to be an algebraic closed field and $f : X \to \mathrm{pic}(\mathbb{S})$. Then the followings are equivalent*

(1) There exists an $\mathbb{E}_\infty$-complex orientation $Mf \to E(L)$.
(2) $L_{K(n)} Mf \neq 0$.

In particular, if we take $X = \mathrm{BU}$ and apply Theorem F, then we will obtain Theorem G.

Remark 4.3. By Hopkins-Smith's nilpotence theorem [DHS88], MU detects nilpotence. That means, $f : R_1 \to R_2$ is nilpotent (in the sense of [HS98][Definition 1]) if and only if $f_* : \mathrm{MU}_* R_1 \to \mathrm{MU}_* R_2$ is nilpotent.

Theorem 4.4 (Strict picard group)**.** *Let L be an algebraic closed field, then we have an equivalence*

$$\underline{\mathrm{Hom}}_{\mathrm{Sp}_{\geqslant 0}}(\mathbb{Z}, \mathrm{pic}(\mathrm{Mod}_{E(L)}\mathrm{Sp}_{T(n)})) \simeq \Sigma^{n+2}\mathbb{Z}_p \oplus \Sigma L^\times$$

References

[BSY22] Robert Burklund, Tomer M Schlank, and Allen Yuan. "The chromatic nullstellensatz". In: *arXiv preprint arXiv:2207.09929* (2022) (cit. on p. 1).

[DHS88] Ethan S Devinatz, Michael J Hopkins, and Jeffrey H Smith. "Nilpotence and stable homotopy theory I". In: *Annals of Mathematics* 128.2 (1988), pp. 207–241 (cit. on p. 3).

[GH04] Paul G Goerss and Michael J Hopkins. "Moduli spaces of commutative ring spectra". In: *Structured ring spectra* 315.151-200 (2004), p. 23 (cit. on p. 1).

[Hah16] Jeremy Hahn. "On the Bousfield classes of H_∞-ring spectra". In: *arXiv preprint arXiv:1612.04386* (2016) (cit. on p. 2).

[HS98] Michael J Hopkins and Jeffrey H Smith. "Nilpotence and stable homotopy theory II". In: *Annals of mathematics* 148.1 (1998), pp. 1–49 (cit. on p. 3).

[Lur25] Jacob Lurie. *Chromatic homotopy theory*. Available online, 2025. URL: https://people.math.harvard.edu/~lurie/252x.html (cit. on p. 1).

[Rez98] Charles Rezk. "Notes on the Hopkins-Miller theorem". In: *Contemporary Mathematics* 220.313-366 (1998), p. 23 (cit. on p. 1).

[Yua24] Allen Yuan. "Examples of chromatic redshift in algebraic K-theory". In: *Journal of the European Mathematical Society* (2024) (cit. on p. 2).